Brightlearn exercise 4: Tshovhole Zwivhuya

Section 1 - UNION

Question 1 * (Combine the tables into one)

```
SELECT    account_id, account_holder, city
FROM      branch_sandton_accounts
UNION
SELECT    account_id, account_holder, city
FROM      branch_rosebank_accounts;
```

Results output

account_id	account_holder
A001	Homvula Dlamini
A002	David Mokoena
A003	Lerato Sithole
A004	Peter Nkosi
A005	Zanele Khumalo
A006	Thabo Motha

Question 2

* We must build a product table catalogue showing product the bank offers without repeats of the product

```
SELECT product_code,
       product_name,
       Product_type
FROM   savings_products
UNION
SELECT product_code,
       Product_name
FROM   current_products;
```

Results table

product_code	product_name
CR01	Standard Current
CR02	Gold current
CR03	Business Current
SV01	Basic Savings
SV02	Premium Savings
SV03	Youth Savings
SV04	Business Savings

Question 3

- Unique list for staff working in either Retail Banking or Cooperate Banking (staff_id, staff_name, email)

```
SELECT   staff_id, email, staff_name
FROM     retail_banking_staff
UNION
SELECT   staff_id, staff_name, email
FROM     Corporate_banking_staff
```

Results table:

staff_id	staff_name
S001	Mpho Radebe
S002	Brian Ishabalala
S003	Aisha Patel
S004	Kabelo Moabelo
S005	Nandi Dube
S006	Sipho Khumalo

Question 4

* Cities , Customers)

expected : city code, city_name, region

```
SELECT    city code, city_name
FROM      mobile_branch_cities
UNION
SELECT    city_code, city_name
FROM      digital_branch_cities,
```

Results table

city_code	city_name
C01	Johannesburg
C02	Pretoria
C03	Cape Town
C04	Durban
C05	Polokwane
C06	Port Elizabeth

Question 5

(customers list)

expected : Customer_id, customer_

```
SELECT    customer_id, customer_name
FROM      push_notification_targets
UNION
SELECT    customer_id, customer_name
FROM      inapp_banner_targets
```

Results table

Customer_id	Customer_name
C1001	Nomsa Zwane
C1002	Andile Buthelezi
C1003	Fatima Mahomed
C1004	Ryno van Zyl
C1005	Thandeka Cele
C1006	Samuel Nkosi

~~Question 6~~

Section 2 : Question 1

(UNION ALL)

Expected outcomes/colums : transaction_id, account_id, amount, transaction_date

```
SELECT   txn_id, account_id, amount
FROM     atm01_transactions
UNION ALL
SELECT   txn_id, account_id, amount
From     atm02_transactions
```

Results table

~~Tra~~ transaction_id	account_id	amount
T1001	A001	500.00
T1002	A002	12000
T1003	A001	300.00
T004	A003	750.00
T1003	A001	300.00
T1005	A004	900.00
T1006	A002	450.00
T1007	A005	150.00

Question 7:

Expected outcome: application_id, customer_id, loan_type, amount requested

```
SELECT    app_id, customer_id, amount_requested)
FROM      gauteng_loan_applications
UNION ALL
SELECT    app_id, customer_id, amount_requested)
FROM      western_cape_loan_application
```

app_id	customer_id	loan_type
LA001	C1001	Home Loan
LA002	C1002	Vehicle Loan
LA003	C1003	Personal
LA004	C1004	Home Loan
LA003	C1003	Personal
LA005	C1005	Vehicle Loan
LA006	C1006	Business
LA007	C1007	Home Loan

Question 8:

expected columns: Complaint_id, customer_id, Category, logged_date

```
SELECT    complaint_id, customer_id,
FROM      email_complaints
UNION ALL
SELECT    complaint_id, customer id,
FROM      app_complaints
```

Results table

complaint_id	customer_id
EC001	C2009
EC002	C2002
EC003	C2003
EC004	C2004
AC001	C2005
AC002	C2001
AC003	C2006
AC004	

9. Question 9

April and May payments records

Outcome: payment_id, account_id, amount, payment_date

```
SELECT  complaint_id, customer id
FROM    email_complaints
UNION ALL
SELECT  complaint_id, customer_id
FROM    app_complaints
```

Results table

Payment_id	account_id	amount
PAY001	A001	12500.00
PAY002	A002	4800.00
PAY003	A003	9200.00
PAY004	A004	3300.00
PAY005	A001	12500.00
PAY006	A005	7600.00
PAY007	A002	5100.00
PAY008	A006	2800.00

Question 10

* Combine debit and credits entries into one general ledger for the audit

expected outcomes: entry_id, account_id, entry_type, amount, entry_date

Results table :
```
SELECT entry_id, account_id, entry_type
FROM debit_entries
UNION ALL
SELECT entry_id, account_id, entry_type
FROM credit_entries
```

entry_id	account_id	entry_type
DR001	A001	Debit
DR002	A002	Debit
DR003	A003	Debit
DR004	A004	Debit
CR001	A001	Credit
CR002	A005	credit
CR003	A002	Credit
CR004	A006	Credit

Bonus questions

Bonus 01 - UNION
Because the fraud analyst only wants a UNIQUE list of customers who appeared on the watchlist and since UNION removes duplicates, a customer appearing on both list will be listed once

Bonus 2
UNION all
Because it will keep all the recorded transactions including duplicate. UNION All keeps all the records.

BONUS 3

Because this is a query involving UNION, the return columns should be equal and since sql returns only 2 columns and the other one returns 3, the new query should return 2, adding

```
SELECT account_id, account_holder
FROM branch_sandton_accounts
UNION
SELECT account_id, account_holder
FROM branch_rosebank_accounts
```